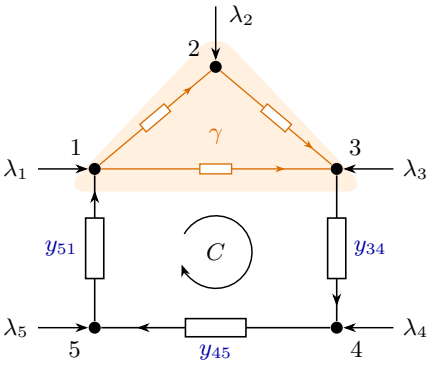
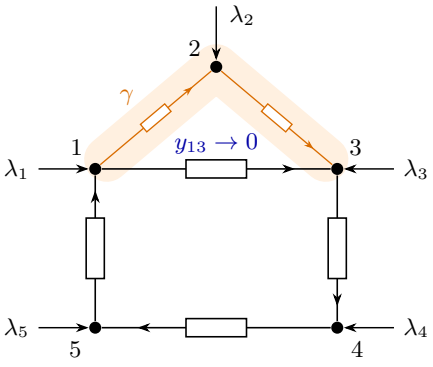
(a) Voltage law of Γ/γ



$$s\,r_{13}y_{13} + r_{34}y_{34} + r_{45}y_{45} + r_{51}y_{51} = 0$$

$$s = 0\text{:}\quad r_{34}y_{34} + r_{45}y_{45} + r_{51}y_{51} = 0$$

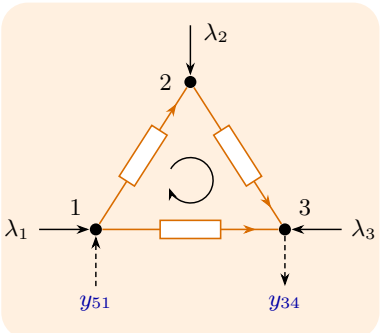
(b) A bypassed resistor: $\gamma = \{12, 23\}$



13 is a self-loop of Γ/γ

$$r_{13}y_{13} = s\,(r_{12}y_{12} + r_{23}y_{23}) \to 0$$

(c) The island, fed by λ^γ



$$\lambda^\gamma_1 = \lambda_1 + y_{51},\quad \lambda^\gamma_2 = \lambda_2,\quad \lambda^\gamma_3 = \lambda_3 - y_{34}$$

$$\textstyle\sum_{v\in S} \lambda^\gamma_v = \lambda_1 + \lambda_2 + \lambda_3 + y_{51} - y_{34} = 0$$

$$r_{12}y_{12} + r_{23}y_{23} - r_{13}y_{13} = 0$$